

Master Thesis

Enhancing Relational Understanding in CLIP
Leveraging Hard Negative Captions (HNC)

Wen Wen

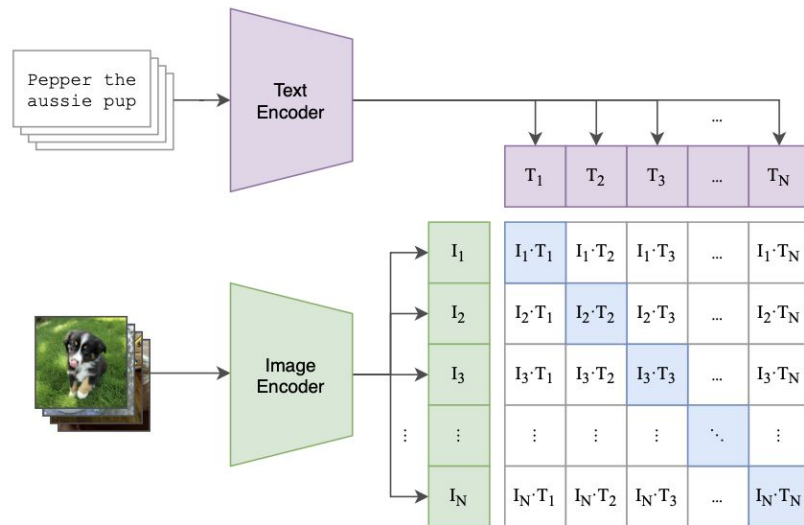
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Prof. Dr. Ngoc Thang Vu

Outline

- Background Knowledge
- Research Question
- Method
- Experiments
- Results and Analysis
- Visual Interpretability
- Conclusions

Background Knowledge

- **CLIP** (Contrastive Language–Image Pretraining)
- **Dual-encoder** model
- Using a **contrastive loss**
- Reward the correct image–caption pairing and penalize all other pairings in the batch
- Strong **Zero-shot** ability
- **Global** embedding alignment



Research Question

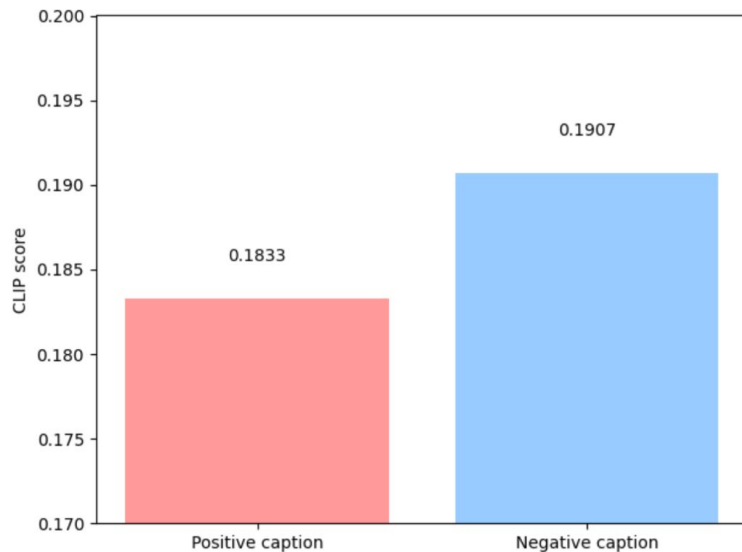
CILP cannot handle relations in images well.



The trees are **behind** the fence



The trees are **in front of** the fence

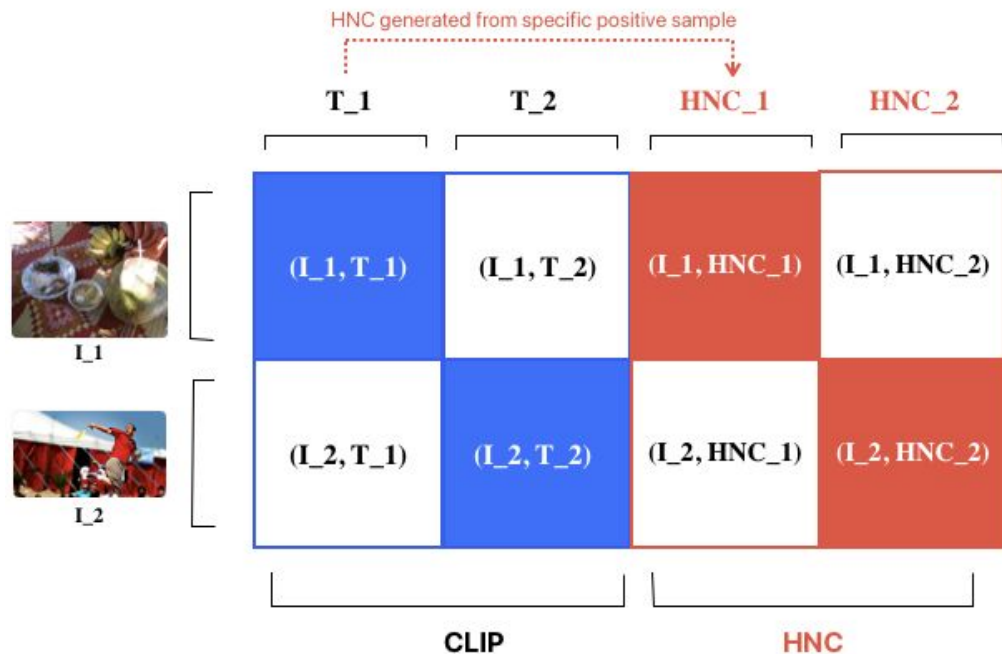


Goal: Improve relational sensitivity

Method:

→ **Inject local information** for global embedding alignment

By combining contrastive learning with **HNC**



HNC example:

Pos: The bear is brown.

HNC: The bear is black.

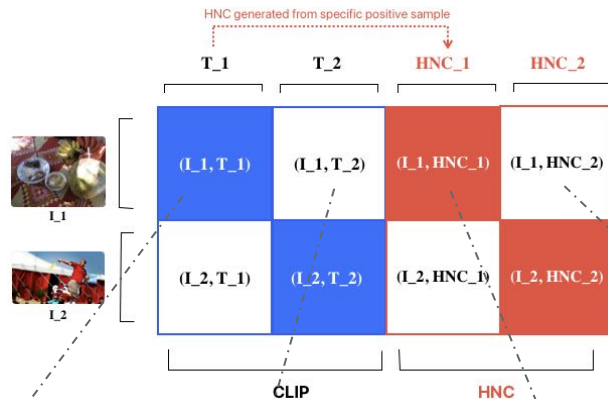
Blue part:

Standard Contrastive learning

Red Part:

Additional HNC part

Loss Function



$$\mathcal{L}_i = -\frac{1}{N} \sum_{i=1}^N \log \left(\exp \left(\frac{\text{sim}(I_i, T_i)}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(I_i, T_j)}{\tau} \right) + \alpha \exp \left(\frac{\text{sim}(I_i, T_i^{\text{HNC}})}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(I_i, T_j^{\text{HNC}})}{\tau} \right) \right)$$

$$\mathcal{L}_t = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp \left(\frac{\text{sim}(T_i, I_i)}{\tau} \right)}{\exp \left(\frac{\text{sim}(T_i, I_i)}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(T_i, I_j)}{\tau} \right)}$$

$$\mathcal{L}_{\text{total}} = \frac{1}{2} (\mathcal{L}_i + \mathcal{L}_t)$$

Regularization Against Catastrophic Forgetting

1. L2 regularization

$$\text{Reg}_{L2} = \sum_i (\theta_i - \theta_i^{(CLIP)})^2$$

2. KL divergence

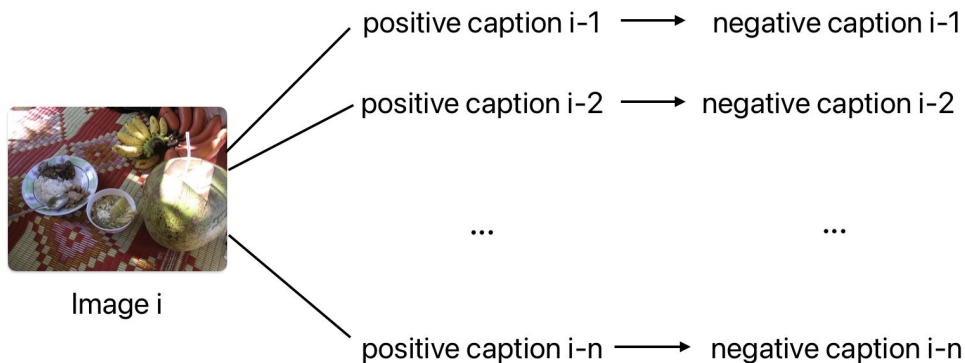
$$s_j = \frac{\exp(s_j/T)}{\sum_k \exp(s_k/T)}, \quad t_j = \frac{\exp(t_j/T)}{\sum_k \exp(t_k/T)}, \quad D_{\text{KL}}(t \parallel s) = \sum_{j=1}^{2B} t_j \log \frac{t_j}{s_j},$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda \cdot \text{Reg}$$

Experiments: Datasets

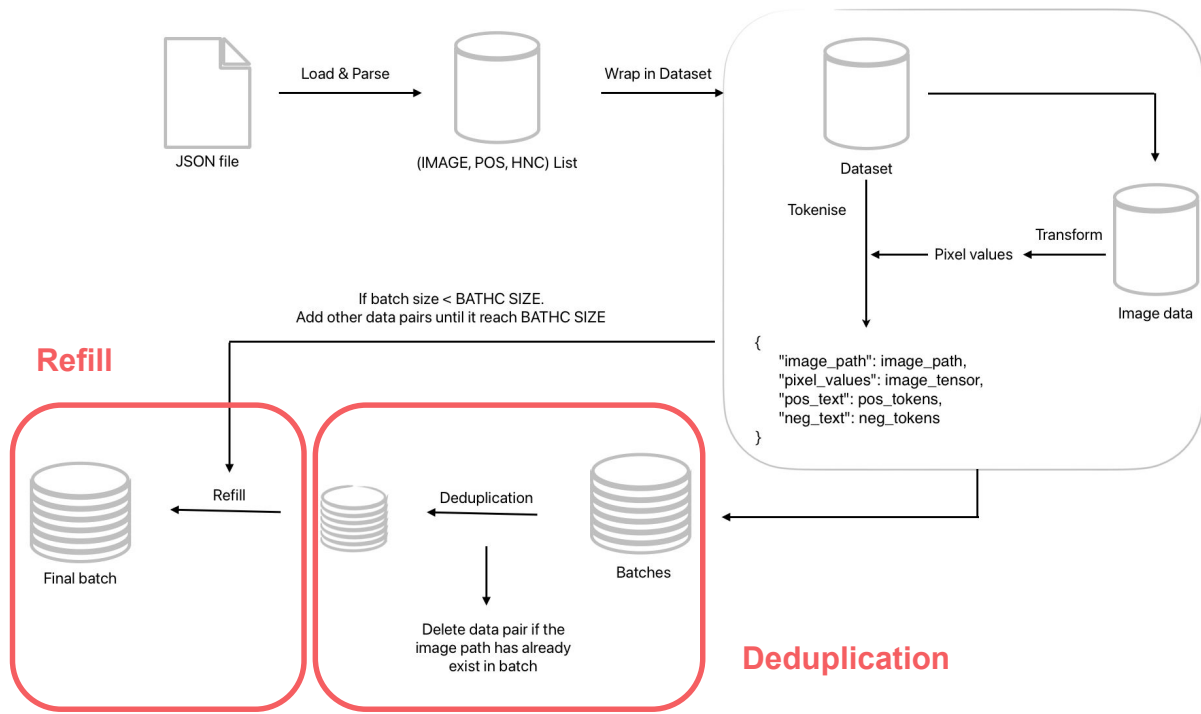
- Training: 100k samples from HNC training dataset.
- Validation: 9k samples from HNC validation dataset.
- Testing:
 - HNC human-annotated set.
 - COCO-derived relational caption pairs.

Data Preprocessing



- One image has **multiple positive captions** and relative hard negative captions.
- For contrastive learning, it's important to remove the repeated image path in each batch.
- Otherwise, repeated positive captions can be treated as random negative captions for that image to influence training

Data Preprocessing



To enforce image uniqueness within each training batch

HNC human-annotated set

- Human-annotated
- **No hard mapping** for positive and negative captions

EXAMPLE:

"n6309.jpg":

{

"0": {"caption": "The people are behind the street", "type": "relation", "label": 0, "replaced": "street, car"},

"1": {"caption": "The people are behind the car", "type": "relation", "label": 1, "replaced": null},

"2": {"caption": "The horses are behind the trees", "type": "relation", "label": 0, "replaced": "behind, in front of"},

"3": {"caption": "The horses are in front of the trees", "type": "relation", "label": 1, "replaced": null},

"4": {"caption": "The car is riding the horses.", "type": "relation", "label": 0, "replaced": "men, car"},

"5": {"caption": "The men are riding the horses.", "type": "relation", "label": 1, "replaced": null}

}

- Problem: Not always following {0, 1, 0, 1, 0, 1... }

HNC Test Dataset Pairing Strategy

- Grouped into consecutive pairs (i and $i+1$)
- If (0, 0) or (1, 1) then skip this pair
- If 'type' is different, then skip this pair
- Select label=1 as positive in each pair
- Humanly check

Test dataset: 950 samples-->After filtering : **872 samples**

Evaluation Metrics

Positive Score:

- Sim(Positive capture, Image)

HNC Score:

- Sim(Hard negative caption, Image)

Margin:

- Sim(Positive capture, Image) - Sim(Hard negative caption, Image)

Random Negative Score

- Sim(Random negative, Image)

Threshold-based accuracy:

$$\frac{s_{\text{pos}}}{s_{\text{neg}}} > \tau$$

- To quantify this contrastive separation
- E.g. Acc@1 means: Sim(Positive, Image)/Sim(HNC, Image)>1

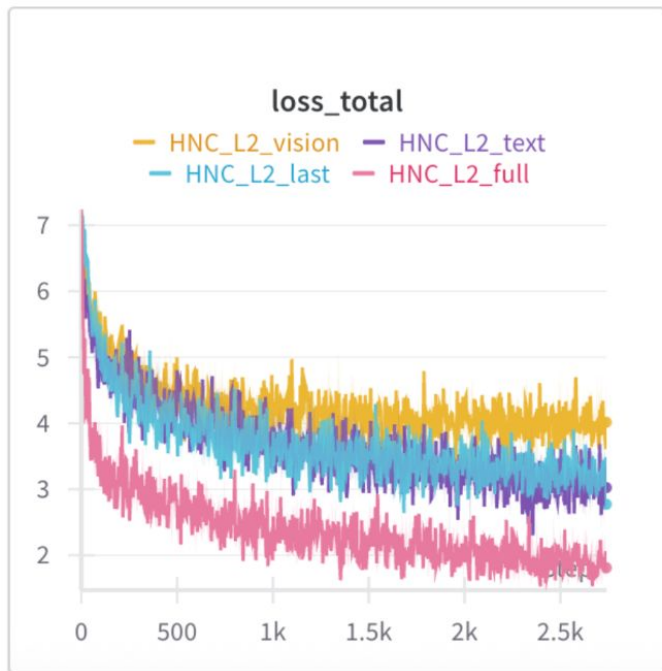
Training parameters

Initialized from the publicly released github CLIP checkpoint.

To control which parts of CLIP are updated during fine-tuning, several parameter selection modes are supported.

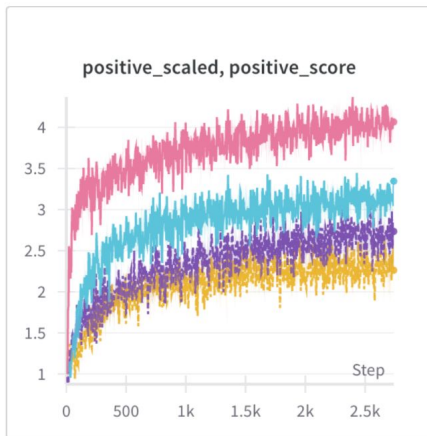
Mode	Text Encoder	Text Projection	Visual Encoder	Vision Projection
text_encoder	✓	✓	—	—
vision_encoder	—	—	✓	✓
full_encoder	✓	✓	✓	✓
last_encoder	last block	—	last block	—

Behavior Across Different Training Parameter Modes

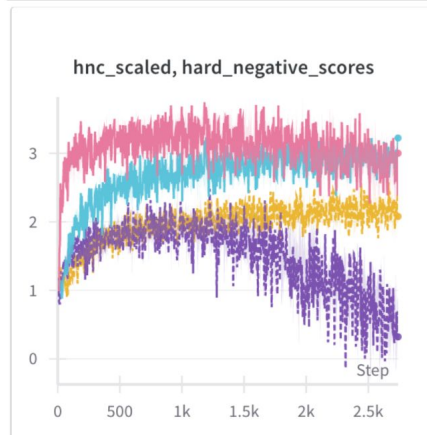


- 20 epoch
- Loss convergence: All configurations are able to fit the training data efficiently in the early stages

Behavior Across Different Training Parameter Modes

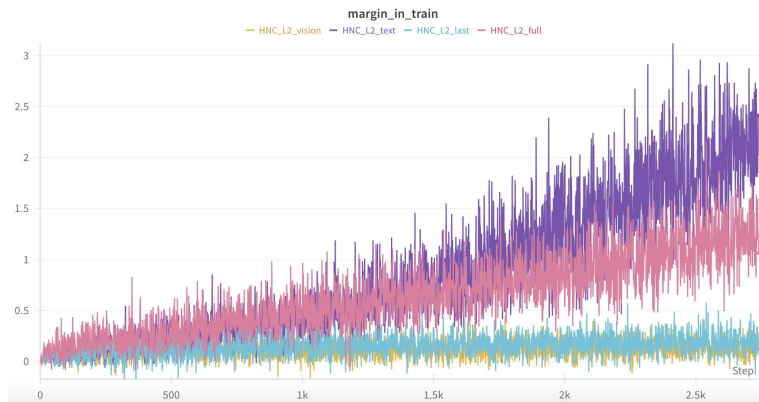


- Positive score get improved
- **Full encoder** configuration showing the largest increase and highest final score



- **Vision encoder** and **last encoder**: increase – level off
 - Models struggle to further suppress semantically similar distractors after an early stage of learning
- **Full encoder** and **text encoder**: increase - decline
 - Growing ability to suppress semantically incorrect captions
- **Text encoder** plays a crucial role, only **visual encoder** provides limited improvement

Behavior Across Different Training Parameter Modes



- Text encoder and full encoder : increase steadily
 - Growing contrastive separation between positive and hard negative captions.
- Vision encoder and last encoder: very limited growth
- Text encoder plays a crucial role

Test scores of different training parameter modes

Each model is selected based on **validation** performance.

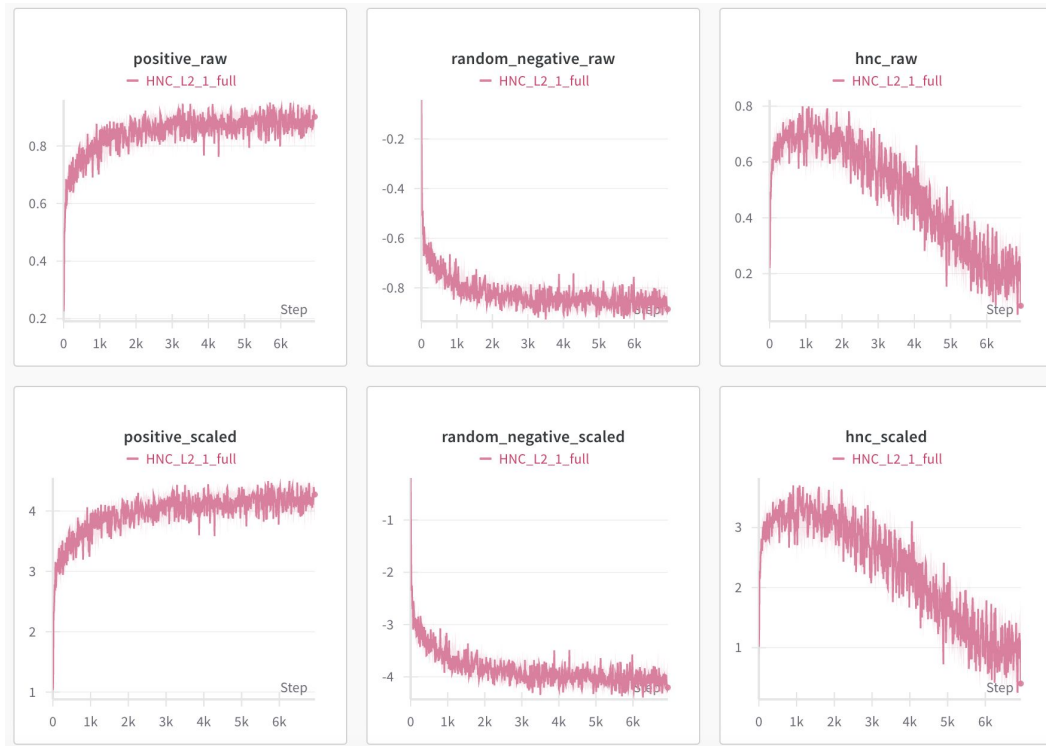
Acc@1 : positive similarity score is greater than or equal to the negative similarity score

In model name: 'L2' means using L2 regularization, 'text' means only the text encoder is updated, as such

Model	Avg_Pos	Avg_Neg	Margin	Rand_Neg	Acc@1
ViT-B/32 (Base)	0.2478	0.2457	0.0021	0.1769	0.5459
HNC_L2_text	0.2448	0.2396	0.0052	-0.0020	0.4587
HNC_L2_vision	0.3552	0.3465	0.0087	0.0035	0.5103
HNC_L2_last	0.4639	0.4515	0.0124	0.0277	0.5264
HNC_L2_full	0.5799	0.5636	0.0163	-0.0028	0.5287

Full encoder performance best: largest positive similarity, largest margin, and lowest random negative score.

Training progression for the full encoder model



During training:

Positive score -> Increased
Random negative score -> Drop
Hard negative score -> Rise then decline

→ Model separates positive and random negative captions then gradually learns to separate positive captions and HNC.

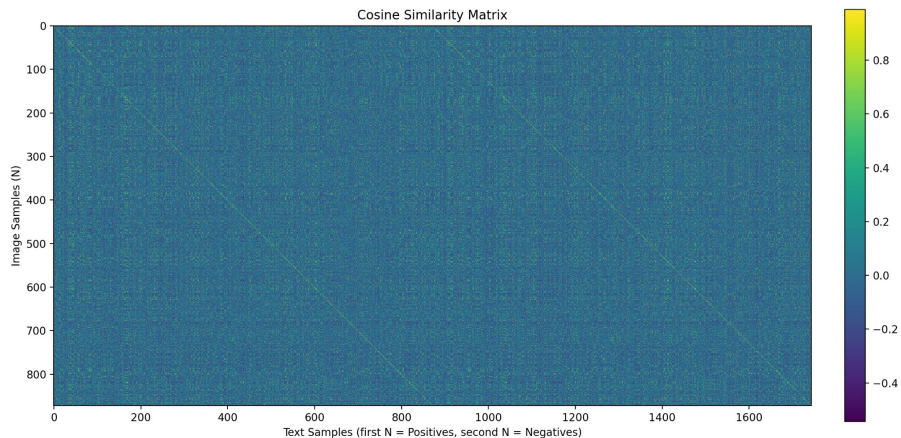
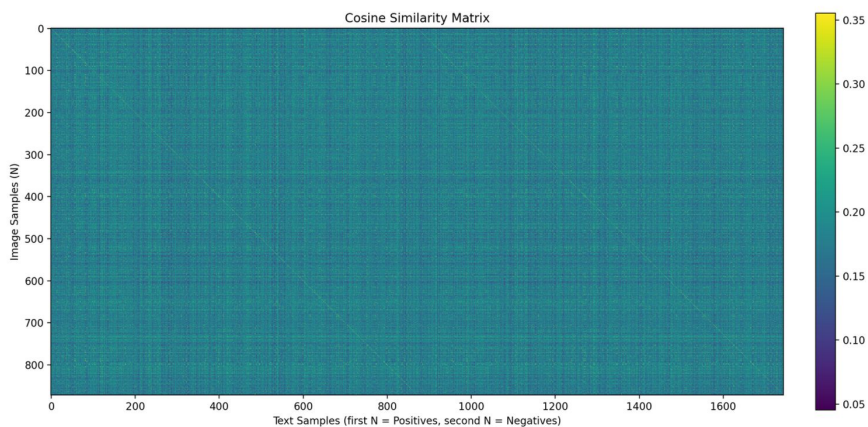
To observe the trend, here running **50 epoch**, 137 step/epoch, $137 \times 50 = 6850$ steps

Cosine Similarity Visualization

Cosine similarity score of **image** and (**positive + hard negative caption**)

Base model (left): cosine similarity scores are relatively low and lack a clear diagonal structure

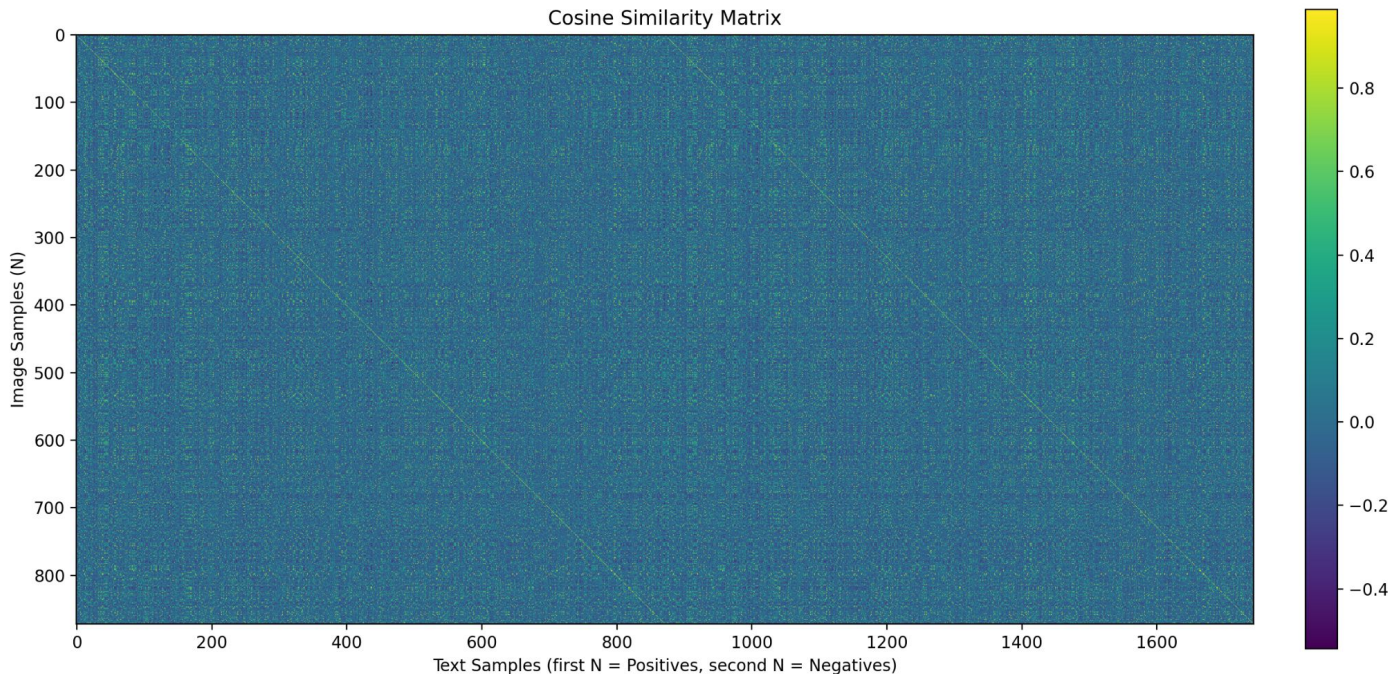
HNC fine tuned model(right): exhibits two diagonal structures



Cosine Similarity Visualization

Left diagonal corresponds to **image-positive caption** pairs, while the right diagonal represents **image-hard negative caption** pairs.

Because the hard negative captions **still contain a lot of image information**. So HNC scores are **higher than random negative** scores.



Comparison of Regularization Strategies: L2 vs. KL

Metric	Base	L2_text	L2_vision	L2_last	L2_full	KL_full
Avg_Pos	0.2478	0.2448	0.3552	0.4639	0.5799	0.6023
Avg_Neg	0.2457	0.2396	0.3465	0.4515	0.5636	0.5779
Margin	0.0021	0.0052	0.0087	0.0124	0.0163	0.0244
Rand_Neg	0.1769	-0.0020	0.0035	0.0277	-0.0028	0.0093
Acc@1	0.5459	0.4587	0.5103	0.5264	0.5287	0.5631
Acc@1.1	0.1227	0.2695	0.2489	0.2592	0.2993	0.2672
Acc@1.2	0.0470	0.1927	0.1755	0.1778	0.2167	0.1927
Acc@1.5	0.0000	0.1261	0.0860	0.0963	0.1227	0.1055
Acc@2	0.0000	0.0849	0.0482	0.0550	0.0711	0.0608
Acc@3	0.0000	0.0493	0.0241	0.0310	0.0333	0.0252

L2: Under stricter semantic threshold(Acc@1.1 and Acc@1.2) performance better

KL divergence: Higher positive scores, higher margin, higher Acc@1 score

- Stronger ability to identify the correct caption among each pair
- Better to prevent catastrophic forgetting in this case

Performance on COCO-Derived Test Set

COCO test dataset: random select 1000 samples

- Average positive score increased
- Margin increased
- Acc@1 decreased

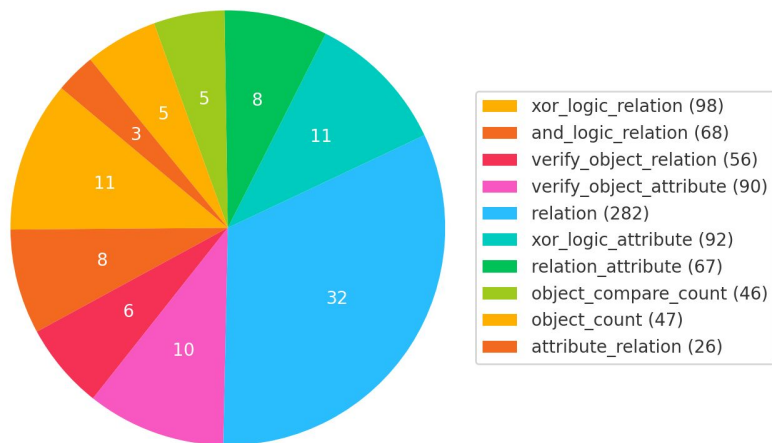
→ Strong generalization beyond the training domain

Metric	Base	L2_full	KL_full
Avg_Pos	0.3043	0.7198	0.7407
Avg_Neg	0.2973	0.7020	0.7255
Margin	0.0071	0.0178	0.0151
Avg_Rand_Neg	0.1510	0.0265	0.0231
Acc@1	0.7160	0.6180	0.6290
Acc@1.1	0.0750	0.1740	0.1170
Acc@1.2	0.0090	0.0930	0.0630
Acc@1.5	0.0000	0.0320	0.0160
Acc@2	0.0000	0.0150	0.0080
Acc@3	0.0000	0.0060	0.0000

Limitations

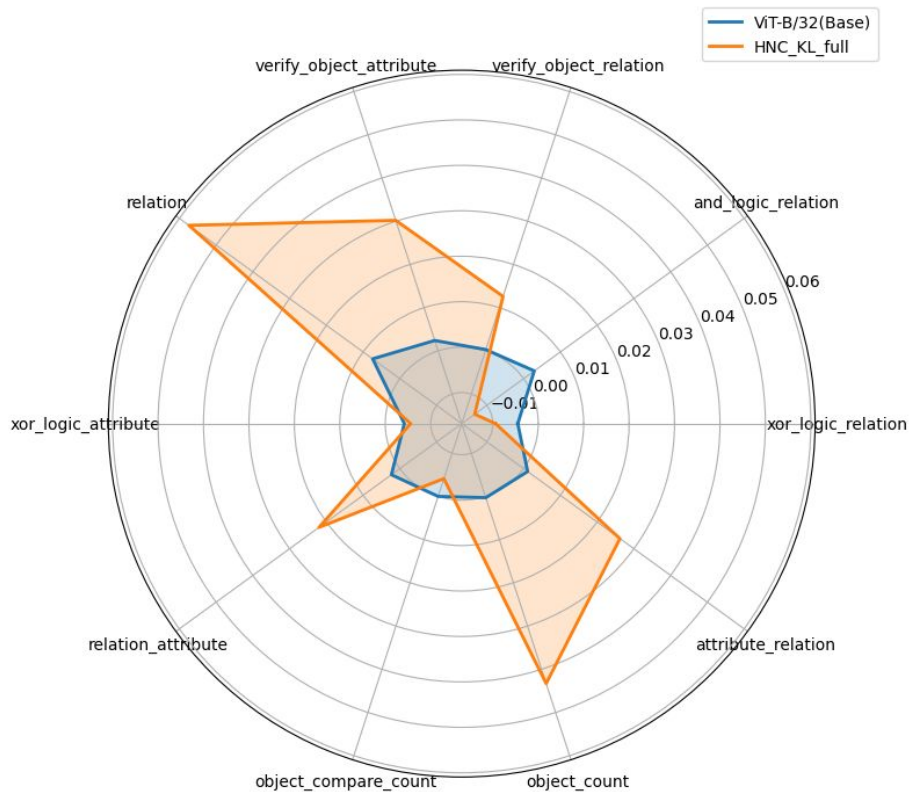
Still struggles with complex logic and coreference.

Performance by semantic type in test dataset:



Type	Samples	ViT-B/32(Base)	HNC_KL_full	Improvement
ALL	872	0.0021	0.0244	0.0223
relation	282	0.0074	0.0574	0.0500
xor_logic_relation	98	-0.0045	-0.0094	-0.0049
xor_logic_attribute	92	-0.0042	-0.0056	-0.0014
verify_object_attribute	90	0.0024	0.0302	0.0278
and_logic_relation	68	0.0029	-0.0133	-0.0162
relation_attribute	67	0.0023	0.0220	0.0197
verify_object_relation	56	0.0003	0.0126	0.0123
object_count	47	0.0003	0.0433	0.0430
object_compare_count	46	0.0000	-0.0041	-0.0041
attribute_relation	26	0.0011	0.0262	0.0251

Comparing margin scores by type



Good at:

Spatial and **object-level** semantics
through contrastive fine-tuning

Bad at:

Logic-intensive categories

- 'and_logic_relation'
- 'xor_logic_relation'
- 'object_compare_count'
- 'xor_logic_attribute'

Examples of Challenging Type

Type - **and_logic_relation**

Pos: There are **both** a lamp **behind** the women **and** a laptop in front of the window.

Neg: There are **both** a lamp **in front of** the women **and** a laptop in front of the window.

Type - **xor_logic_relation**

Pos: The laptop is **either** behind the women **or** the sofa.

Neg: The laptop is **either** behind the women **or** the foot.

Type - **object_compare_count**

Pos: There are **as many** remote controls **as** people.

Neg: There are **fewer** remote controls **than** people.

Type - **xor_logic_attribute**

Pos: There is **either** a red carpet **or** green curtains.

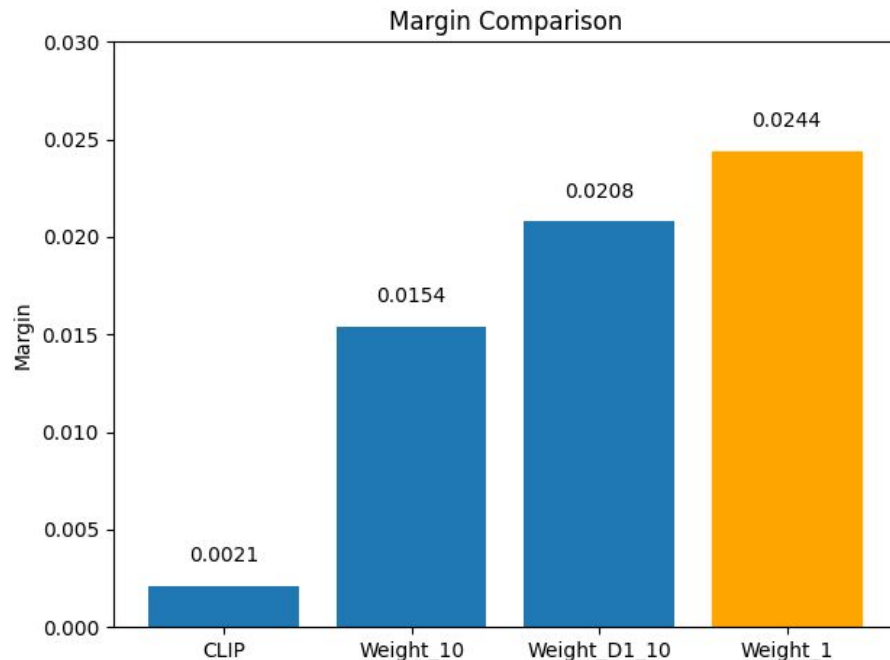
Neg: There is **either** a red carpet **or** purple curtains.



Limitations

Overweighting hard negatives may hurt stability

- Larger HNC weights do not always lead to better performance.
- Using dynamic weight (small at first, increasing the weight during training), better than fixed large weight, but still performance bad.



Visual Interpretability

Based on a second-order attributions method for CLIP-based models (Tilli et al.)

- Decomposes the similarity score into fine-grained contributions from image-text feature pairs
- Captures cross-modal interactions between specific words and regions

Pos: The trees are behind the fence

- text-to-image

Original Image



Base CLIP
score=0.0430



Fine-tuned CLIP
score=0.0504



Thanks to Dr. Pascal Tilli and their team.

<https://arxiv.org/abs/2408.14153>

Visual Interpretability

Pos: The trees are behind the fence

- Token-level image-to-text attribution visualization

Base CLIP

the trees are behind the fence

Fine-tuned CLIP

the trees are behind the fence

Visual Interpretability

Pos: The **trees** are behind the fence.

Base CLIP: "trees"



Fine-tuned CLIP: "trees"



Visual Interpretability

Pos: The trees are behind the **fence**.

Base CLIP: "fence"



Fine-tuned CLIP: "fence"



→ Object detection get improved. E.g. trees, fence

Visual Interpretability

Pos: The trees are **behind** the fence.

Base CLIP: "behind"



Fine-tuned CLIP: "behind"



→ For relational words (e.g. behind) can also focus on the right place in image.

Conclusions

- Relational understanding get improved by injecting HNC in contrastive learning
- Text encoder plays a central role.
- Full encoder tuning achieves the best overall performance.
- KL regularization yields highest margin, can better deal with catastrophic forgetting in this case, while L2 performance better at strict thresholds.
- Good out-of-domain generalization (On a COCO-derived test set).
- Visual interpretability shows improvement.

Thank you

✨ Special Thanks ✨

🌸 Dr. Pascal Tilli 🌸

🌻 Prof. Dr. Ngoc Thang Vu 🌻

🌺 Weekly Thesis Meeting 🌺